

ISO 9001
ISO 14001
OHSAS 18001



• Technology • Innovation • Excellence

Liquid Ring Compressors for Petrochemical, Nuclear & Chemical Process Plants



.....the next generation technology



Unmatched Technology,
Superior Performance,
Trouble Free Operation



Technology • Innovation • Excellence

COMPANY

Swam a group known for technology, innovation and excellence is the country's largest manufacturers of Pressure and Vacuum Technology Compressors, Blowers & Pumps. The Swam Liquid Ring Compressors are used to handle Mixed Gas, Coke-oven Gas, Wet Gases, Gases with fraction solids, Condensable Toxic, Corrosive and Explosive Gases.

Swam Liquid Ring Technology, developed by our technical highly skilled engineers are of advanced design superior performance & of high reliability. The design and construction of the Liquid Ring Vacuum Pump & Compressors reflect the experience and know-how acquired over years of product development in this particular field of machine engineering. These compressors are used in Refineries and Petrochemical Plants & other Power Plants.

With over three decades of experience the company has adequately matured to execute different projects and is today represented in over 15 countries. Our quality system has been accredited with ISO-9001 quality standards and procedures. The company's head office & works are located at Noida near New Delhi, the capital city of India. The company has FOUR manufacturing plants equipped with CNC Machines & in-house testing facility upto 1000 kw.

Our highly motivated engineers and work force faces, each contract with commitment to meet the client process specification & quality requirement. The company has executed many prestigious projects under the stringent quality compliance.

The company's quality system is as per ISO 9001:2008 and has very high regard for safety and environment and hence has all plants certified to ISO 14001 & OHSAS 18001.



FOCUS ON QUALITY & RELIABILITY

Emphasis on quality and performance is a commitment of every individual at Swam beginning at grass root level. The quality control is practiced at all levels of the organization and all its aspects are thoroughly complied with the specified standard and specification, till the product gets delivered.

The quality assurance system encompasses self quality control by each worker, supervised by quality control engineers. Each product undergoes thorough & stringent tests at different stages of manufacturing and are test run for several hours prior to despatch as per the relevant code, customers' specification & approved quality plan.

ADVANCE MANUFACTURING SYSTEM

Swam has the most advance manufacturing system with a large number of CNC Machines installed in our four plants. The Liquid Ring Compressors and other products are manufactured with most optimum profile resulting higher efficiencies, high displacement, lower slippage. Thus our products are more energy efficient compared to other.

Latest Reliable Technology fo the Industrial Processes

Swam compressors provide lower maintenance and lower life cycle costs and greater overall value to our clients. The Liquid Ring Compressors have improved design, enabling wider ranger of operation, higher efficiency, saving of water and simpler seal line piping.





The following are some of the important aspects while selecting a suitable model and type of Compressor :

- **Application**
- **Media handled**
- **Characteristic of Gas**
- **Suction conditions**
- **Sealing system and M.O.C**
- **Process parameters**

Our quality system has been accredited with ISO-9001 quality standards and procedures, and some of our products are CE certified.

- **Swam** Liquid Ring Compressors have been developed through its extensive experience & technical expertise. The series, feature high performance, low maintenance and embody the state-of-the-art design.
- The Compressors have been standardized, with a wide applicable working pressures. Ranging from low to high & can be effectively used for gas conveying, exhausting and boosting applications for positive and negative pressures.

Swam Liquid Ring Compressors are used in Refineries, Petrochemical Plants & other Process Industries.

The Compressors are fitted with special sealing system to make it leak-proof. The sealing system varies depending upon the application, media handled. The construction materials are special, depending upon again the nature of gas.

SWAM LIQUID RING GAS COMPRESSOR

Available upto 6,000 M³/Hr.

CASING

Integral cast body of corrosion resistant material. The casings have large in-built suction and discharge manifolds and heavy cast ribs. Depending on application, the casing can be of Cast Iron, SG Iron, Carbon Steel, Stainless Steel or of Ni. Alloy Cast Iron or Duplex Stainless Steel.

LUBRICATION

For proper lubrication and excellent durability, the bearings located on either end.

Superior Features of Swam Liquid Ring Compressors

- Advanced design.
- The Internals and body are all manufactured on CNC and hence, very high accuracies.
- The sealing system consist of single mechanical seal, double acting Mechanical seal or dry gas seal.
- Compressors are energy efficient due to Unique high efficiency impeller profile.
- Design compliance to API.
- The complete Compressor package is PLC operated with complete instrumentation and controls and logics.
- Complete Compressor package system is leak proof ensuring no leakages at all.
- Rigid design of the casing, strong and rugged in compression.
- Bearings are supported at both ends of the shaft. Radial load is balanced by two cylindrical roller bearings mounted symmetrically at both ends of the shaft. Axial location is through coupled angular contact ball bearings. Rigid structure and stable performance. All bearings are SKF / FAG bearings.
- The impellers are dynamically balanced to grade 2.5 of ISO 1940.
- Rubber O ring is applied for casing sealing, capable of withstanding high pressure.



Wide application, outstanding achievement

Swam Liquid Ring Compressors has the advantages of handling various types of working gas compression and sealing. During the working process, the gas is compressed nearly at constant temperature, very safe, especially suitable for the treatment of hazardous gas, inflammable, explosive, corrosive. In many industries, Liquid Ring Compressors have been irreplaceable by other type of compressors.

Swam Liquid Ring Compressors have obtained many customers' trust for being energy saving, reliable and stable performance etc.

Swam Liquid Ring Compressors are also available for following applications.

- Recovery of atmospheric tower gas and primary tower gas during oil refinery based on atmospheric and vacuum distillation.
- Compression of nitrous acid gas in hexanedioic acid industry.
- Compression of dry chlorine gas and dry hydrogen gas in chloralkali industry.
- Recovery of exhaust gas of flare apparatus in chemical industry.
- Recovery of flue exhaust gas of salt mine.
- Compression of oil gas on offshore oil platform.
- Flare gas exhauster compression.

Swam Liquid Ring Compressors are energy-efficient, safe and reliable and easy to maintain, is Swam's latest developed liquid ring compressor series. Designed & construction in single action type & double action type according to the advanced method of Computational Fluid Dynamics.

Wide Performance Range

Various Models & specifications with performance options at different speeds. Users can find the most suitable product for their application according to the technical data such as discharge pressure, suction capacity etc. The discharge pressure of a single stage compressor can reach 550 KPaG while that of double stage service at much higher uptill 1200 KPaG.

The compressors are available in various models with capacities from 100m³/hr to 6000m³/hr.

SLC- 150	100 – 200 m ³ /hr.
SLC- 200	250 – 400 m ³ /hr.
SLC- 500	500 – 750 m ³ /hr.
SLC- 800	800 – 950 m ³ /hr.
SLC- 1000	1000 – 1200 m ³ /hr.
SLC- 1500	1200 – 2000 m ³ /hr.
SLC- 2000	1800 – 2600 m ³ /hr.
SLC- 3000	2600 – 3500 m ³ /hr.
SLC- 4200	3400 – 4500 m ³ /hr.
SLC- 5200	4500 – 5400 m ³ /hr.
SLC- 6200	5200 – 6100 m ³ /hr.

Advanced Hydrodynamic Model

Swam keeps ahead of applying CFD method to the design of the hydrodynamic model of Liquid Ring Compressor. The compressor optimized through this innovative method is very energy-efficient and brings immediate economical benefits to the users.



REFINERY APPLICATIONS

Vacuum Filtration:

Swam compressors are used extensively for the rotary drum and disc filters used for solvent de-waxing, waste water treatment and additive / lubricant production. Their ability to remove water and handle carryover more efficiently and reliably makes them the ideal choice for these installations.



Vapor Recovery:

Swam compressors are used to recover hydrocarbon vapors vented from storage tanks. The condensate of the vapor can be used as the seal liquid in order to:

- Avoid contamination of the recovered product
- Eliminate waste
- Reduce the pump size by condensing the gas vapors

Swam products are not limited to refineries. They provide substantial benefits for upstream and offshore applications.

Coal bed methane extraction:

The liquid ring pump's ability to be used simultaneously as both a vacuum pump & a compressor make it a great choice for extracting natural gas from coal seams & conveying it to a pipeline compressor house.

Oil field gas boosting:

Swam compressors are used to convey gases, which have been pulled off of oil wells, to storage or processing facilities.

Crude stabilization:

Swam Liquid Ring Pumps are used to remove dissolved gases from crude oil tanks, thus stabilizing the true vapor pressure of the crude. The removed gas can be boosted downstream and used as fuel.

Offshore oil rigs:

Swam systems also provide great performance on offshore rigs, where they are used for gas compression, vapor recovery and glycol recovery systems.



COMPRESSORS FOR OFFSHORE APPLICATIONS

Flare Gas Recovery Vent Gas Recovery Glycol Recovery

Swam is committed helping you meet the stringent environmental policy which prohibits the flaring of gas. Our New Generation Compressors are the ideal choice for recovering fuel gas and condensing valuable hydrocarbons.

When reviewing the total life cycle cost of a flare gas compressor system, the **Swam** Liquid Ring System provides more value & better reliability than any other technology.

Swam Liquid Ring Compressors run cool and they clean the gas as they compress it - eliminating the need for downstream after condensers and scrubbers. The result is simple, more reliable system.

Flare Gas Recovery Compressors

Swam is committed to helping you meet the stringent environmental policy & regulations which prohibits the flaring of waste gas. Our Compressors are the ideal choice for recovering fuel gas and condensing valuable hydrocarbons, making more money for the refinery.

Crude Tower Vacuum System

Whether you are building a new vacuum crude distillation tower or upgrading an existing one, **Swam** can provide the most efficient system. **Swam** Compressors are much more efficient than steam ejector vacuum systems. This is supported by our engineering design expertise, high quality manufacturing practices and the wide range of our compressor models available for the service.

The annual cost of operating a tower vacuum system will be several times the original equipment cost. It is imperative that we recognize that most companies keep a tight rein on capital investments, whether new systems or cost saving upgrades - and **Swam** can offer you various system options to help you strike the optimum balance between capital costs and operational savings.



MATERIAL OF CONSTRUCTION

Swan has a large range for material of construction from, Cast Iron, Cast Steel, Stainless Steel, Duplex Steel etc.

Options of material of wet parts of Swan Compressors

Part Name	Material and Code					
Casing	Cast Steel	304	304 L	316	316 L	Duplex SS
Impeller	Cast Steel	CF-8	CF-3	CF-8M	CF-3M	Duplex SS
Shaft	SS 410	SS 410	SS 410	SS 410	SS 410	Duplex SS
End Shield	Cast Steel	CF-8	CF-3	CF-8M	CF-3M	Duplex SS

Swan Compressors packages and automation control

Swan is very experienced in manufacturing and packaging electro-mechanical integration equipment package. The compressor equipment package can be automatically controlled in varied ways, adaptable to multiple application conditions and greatly reducing the operational cost, surely the most favorable product for the users. The compressor has wide applied application in the petroleum processing and chemical processing industry.





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Swam Vacuum Compressors for the naphtha / solvent recovery process applications

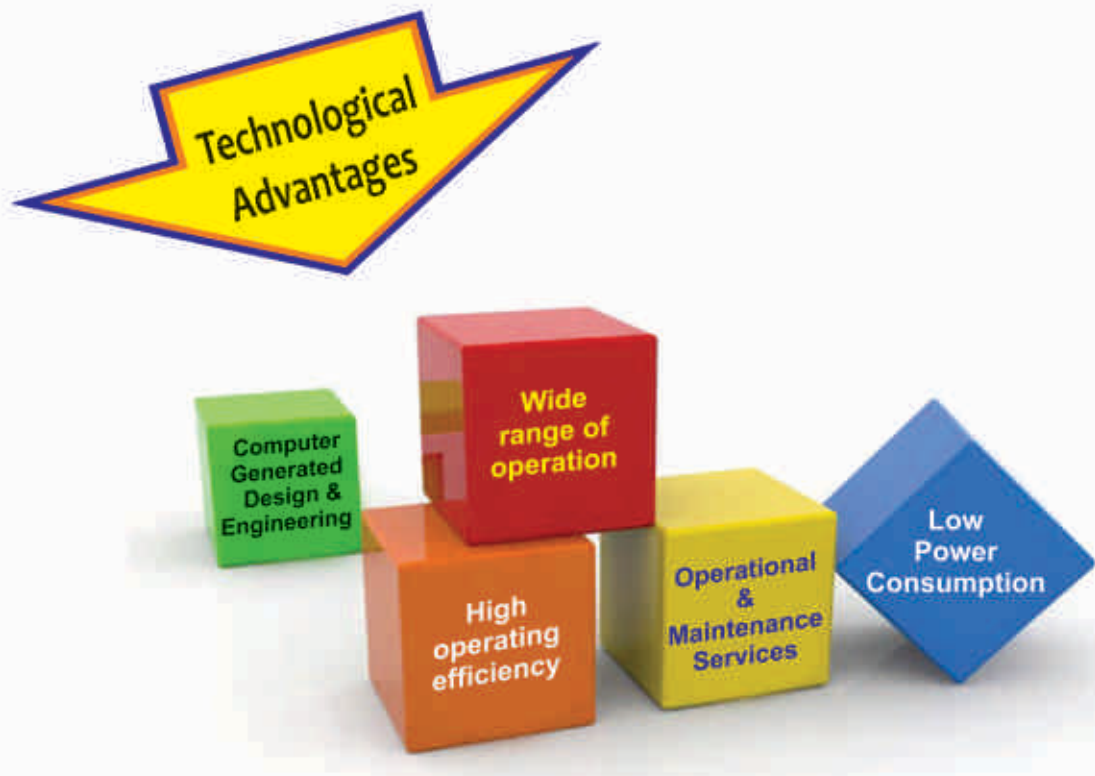
Cool running **Swam** vacuum compressors have the unique ability of acting as a condenser when extracting a saturated vapor from a distillation process. In addition, many solvent can be used as the seal liquid - including naphtha - allowing you to seal the vacuum compressor with the same liquid that is being condensed. The result is a vacuum pump system that reclaims valuable solvents without contamination. The recovered solvent can then be reused in the process or sold.

An advanced proven technology

Swam has been providing the most reliable compressors designed for all requirements. Swam is the industry standard for flare gas recovery, vent gas boosting, vapor recovery (vacuum) and crude tower vacuum systems and many other critical processes even for Nuclear plants.



Technological Advantages



Why SWAM as best Choice ...

- ISO-9001 Quality certified with ISO 14001 & OHSAS 18001
- Experience of over three decades
- An installation of over 50000 units world wide
- Presence in over 12 countries
- Quality Focus Organisation
- Strong Project Management team
- Innovation in Design & Development
- Team of highly qualified professionals
- Competent team for Customer support, Operational & Maintenance
- Advanced technology products compiling to international standards



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Customer Support Services :

A team of experienced and qualified engineers are deployed to rendering all kind of technical post supply services to our esteemed customers; such as pre-commissioning and post commissioning services, trouble shooting etc., to make the equipment perform to user's / clients satisfaction.

Professionally and technically competent personnel are always ready to render these services, promptly and expeditiously.

For details contact H. O. or the nearest service network.

Manufacturing Plants:



PLANT - I



PLANT - II



PLANT - III



SWAM PNEUMATICS PVT. LTD.

(ISO - 9001 : 2008 Quality Certified)

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For details of our local contact office in your region, contact H.O. or log on to website



Leader in : Air-Gas Conveying, Boosting & Vacuum Technology